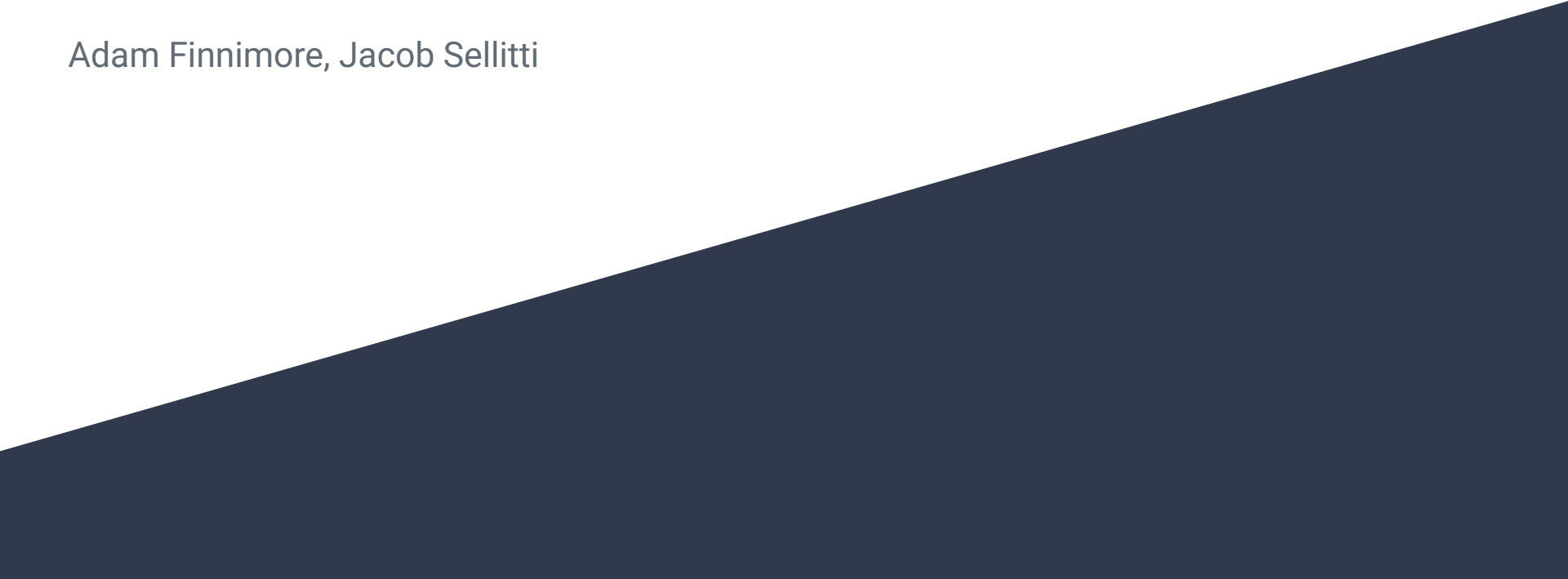


Predicting Expected Batting Average as a Measure of Hitting Success in the MLB

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A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.

Intro and Motivation

Predicting hitting success in baseball is a difficult task, largely because there is no one measure of a batter's skill.

Establishing a standardized metric for hitting proficiency, then training a model to predict it based on hitting stats, would be very useful.

Even though baseball is the most datafied sport in the world, especially with the implementation of MLB's Statcast system in 2015, this is incredibly complicated.



Intro and Motivation

Expected Batting Average (xBA) can potentially be used for this purpose as a measure of the likelihood of a batted ball resulting in a hit.

This stat is already computed through Statcast which compares the ball's launch speed and exit velocity to past hits and uses some type of KNN/Clustering algorithm to predict a hit's likelihood (the actual formula is unknown)

Our goal is to build a model to predict how likely a ball is to be a hit based on previous data (xBA), and to compare this to traditional hitting metrics to model hitting and overall season success for players and teams.

Data

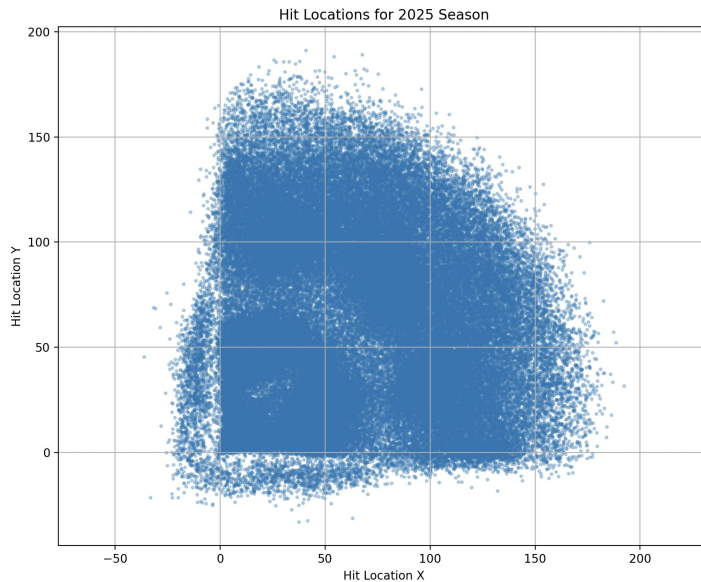
We are using the BaseballReference and Baseball Savant datasets, which contain data for every pitch from the 2025 MLB season. By scraping this with the pybaseball library, we have 118 variables, including pitcher and batter ID, pitch type, the result of the pitch, larger game and context data, etc.

This gave us 183,092 baseball events.

game_type	stand	p_throws	home_team	away_team	type	hit_location	bb_type	balls	strikes	game_year	pb_x	pb_z	plate_x	plate_z	on_3b	on_2b	on_1b	outs_when_ininning	inning_1	
R	L	R	BOS	DET	X	7 fly_ball		0	0	2025	-0.71	0.00	-0.3958864	0.0631750202025187		700242	628451	2	9 Top	
R	R	R	BOS	DET	S	2		0	2	2025	-0.80	0.93	0.67640306	0.03139198670988		700242	628451	1	9 Top	
R	R	R	BOS	DET	X	6 ground_ball		2	2	2025	-1.47	0.28	0.736510870	1.984433703956037			700242	1	9 Top	
R	L	R	BOS	DET	B	3		2	2025	-0.85	1.08	1.195847480	1.1866080304996394				1	9 Top		
R	R	R	BOS	DET	S	2		0	2	2025	1.73	0.68	0.295167877	1.01211578510867				0	9 Top	
R	L	R	BOS	DET	X	3 ground_ball		1	0	2025	-0.67	1.5	0.705668631	1.9977820591629691				2	8 Bot	
R	L	R	BOS	DET	X	4 ground_ball		2	2	2025	-0.03	-0.21	-0.3721082	2.78258302903691				1	8 Bot	
R	L	R	BOS	DET	S	2		2	2	2025	0.24	0	0.966016300	0.9602011754936126				0	8 Bot	
R	L	R	BOS	DET	X	3 ground_ball		0	2025	0.21	0.68	0.14030971	0.48823980509347				2	8 Bot		
R	L	R	BOS	DET	X	9 fly_ball		1	1	2025	0.22	0.48	-0.111480	0.45397974445034				1	8 Bot	
R	R	R	BOS	DET	X	5 ground_ball		1	0	2025	-0.13	0.71	-0.3113563	0.7676309172311095				0	8 Bot	
R	R	R	BOS	DET	S	2		0	2	2025	-0.53	1.47	0.176048730	0.495716832211095				678882	1	7 Bot
R	L	R	BOS	DET	X	8 line_drive		0	1	2025	-0.62	1.45	0.0319076	1.146054024498323				678882	1	7 Bot
R	L	R	BOS	DET	X	5 popup		0	0	2025	-0.62	1.59	0.228216410	1.144812796415184				678882	0	7 Bot
R	R	R	BOS	DET	B	3		2	2025	1.16	-0.02	-0.1917628	1.20186386	700242			678009	2	7 Top	
R	L	R	BOS	DET	B	3		3	1	2025	-0.54	1.17	-0.6915853	0.799244141	700242				2	7 Top
R	R	R	BOS	DET	X	5 ground_ball		1	2	2025	1.22	-0.46	0.100625437	0.81664818		700242	628451	0	7 Top	
R	R	R	BOS	DET	X	8 line_drive		0	1	2025	-1.47	1.45	0.1445661	2.263509		700242	0	7 Top		
R	L	R	BOS	DET	B	3		2	2025	-0.66	1.15	0.65431190	0.477602983998004				0	7 Top		
R	L	R	BOS	DET	X	7 fly_ball		2	0	2025	-1.26	0.97	-0.9367591	0.3236329710141383				2	6 Bot	
R	R	R	BOS	DET	X	7 fly_ball		1	2	2025	0.39	0.59	-0.2202738	0.0000124778641903				1	6 Bot	
R	L	R	BOS	DET	X	8 fly_ball		0	0	2025	-1.11	0.96	0.25066291	0.26613328775597				0	6 Bot	
R	R	R	BOS	DET	S	2		2	2	2025	-0.49	1.33	0.693735790	1.8263601480041487				2	6 Top	
R	L	R	BOS	DET	X	9 fly_ball		1	1	2025	-0.66	1.16	0.458249600	0.2091128014644954				1	6 Top	
R	L	R	BOS	DET	S	2		1	2	2025	1.25	-0.72	-0.4029052	0.37749819223443				0	6 Top	
R	L	R	BOS	DET	S	2		0	2	2025	0.64	0.29	-0.3996218	1.773302744	677800	663993	657136	2	5 Bot	
R	L	R	BOS	DET	X	6 popup		2	2	2025	1.18	0.39	0.484090652	0.285541688	677800	663993	657136	1	5 Bot	
R	R	R	BOS	DET	B	3		3	2	2025	-1.17	1.04	1.136242897	2.290996308873857		677800	663993	1	5 Bot	
R	L	R	BOS	DET	B	3		3	2	2025	-1.36	0.11	0.221764740	0.4244391817457101		677800	677800	1	5 Bot	
R	L	R	BOS	DET	B	3		2	2025	-1	1.49	0.664381985	0.8298930019553537				1	5 Bot		
R	R	R	BOS	DET	X	9 fly_ball		2	1	2025	-0.73	1.47	0.348157397	0.798938985676183				0	5 Bot	
R	R	R	BOS	DET	S	2		0	2	2025	1.24	-0.35	0.61961881	1.2381489953886151				2	5 Top	
R	L	R	BOS	DET	X	8 fly_ball		0	0	2025	-0.79	1.08	-0.4322175	0.484217885917777				1	5 Top	
R	L	R	BOS	DET	X	7 line_drive		1	1	2025	0.5	0.22	-0.5480	0.764427800475884				0	5 Top	
R	L	R	BOS	DET	X	6 popup		2	2	2025	-0.67	1.47	0.59605591	0.024411977689985		680776		2	4 Bot	
R	R	R	BOS	DET	S	3		0	3	2025	-0.68	1.41	0.41786384	0.2637020249811512		680776		1	4 Bot	

Data

Data cleaning steps include filtering by only pitches that resulted in a ball in play (i.e. that have a valid xBA value), and standardizing hit location coordinates to better visualize certain features.



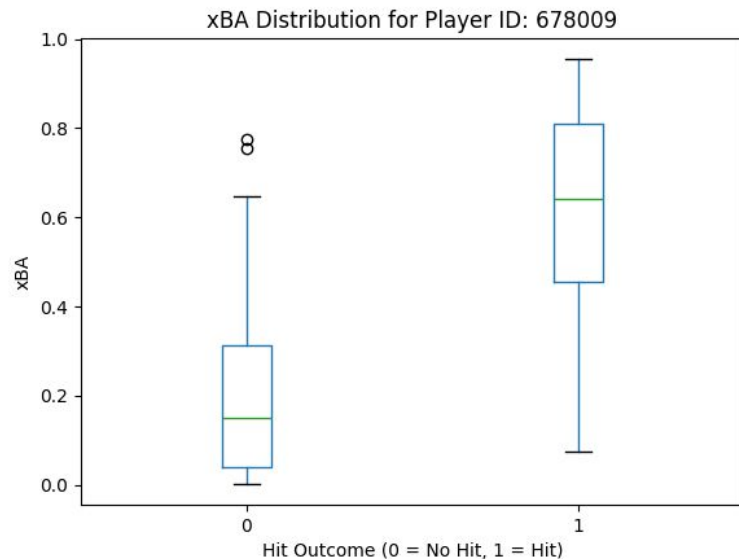
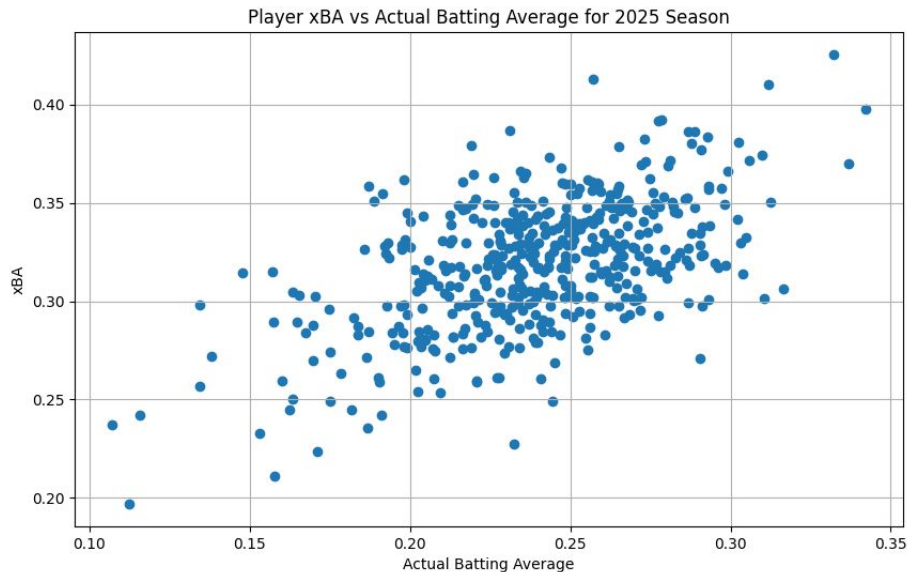
Eventually, our final feature set included 122,071 observations with 7 input values as well as some metadata and labeling features.

Codebook

Variable Name	Description	Type	Values / Notes
game_pk	Unique identifier for each game	Categorical	6 digit ID code
batter	Unique identifier for each batter	Categorical	6 digit ID code
pitcher	Unique identifier for each pitcher	Categorical	6 digit ID code
launch_speed	Speed of the ball off of the bat	Numerical	Measured in miles per hour
launch_angle	Vertical angle of the ball off of the bat	Numerical	Measured in degrees; negative = downward, positive = upward
hit_distance	Distance or estimated distance ball traveled in air before hitting the ground	Numerical	Measured in feet; estimated for home runs that landed in the stands
hit_location_x	x coordinate of the location where the ball first landed	Numerical	location; used to estimate direction that the ball was hit
hit_location_y	y coordinate of the location where the ball first landed	Numerical	location; used to estimate direction that the ball was hit
pitch_speed	Speed of the pitch at release	Numerical	Measured in mph
release_spin_rate	Rotation of the ball at release	Numerical	Measured in rotations per minute (rpm)
xBA	Expected batting average	Numerical (proportion)	Range 0-1; Statcast metric that measures the probability that a ball with the given launch speed and angle will result in a hit
events	Result of the play	Categorical	single, double, triple, home_run, field_out, force_out, grounded_into_double_play, etc
team	Batting team name	Categorical	3-letter MLB team name abbreviations (e.g., BOS = Boston Red Sox, NYY = New York Yankees)
hit	Records if a hit occurs	Binary	0-1; 0 = out, 1 = hit

EDA Highlights

We can visualize the relationship/correlation between batting average and expected batting average to see which players may have been the luckiest vs. unluckiest (filtered by players that had more than 50 PA).

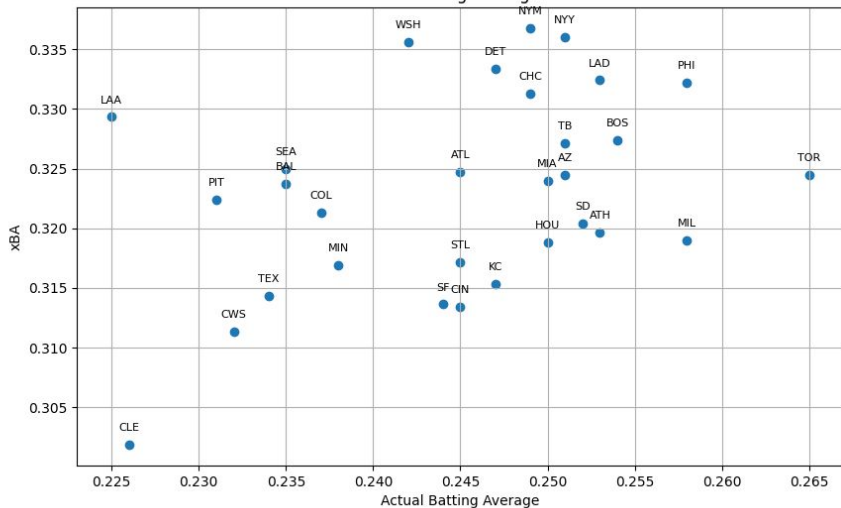


Sample distribution of xBA values for every ball hit by Aaron Judge in the 2025 season

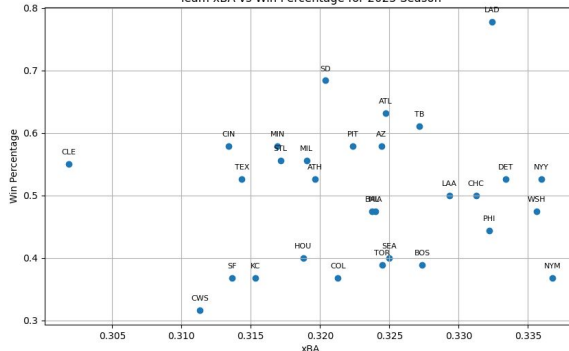
EDA Highlights

This can also be done at the team level to interpret team hitting efficiency vs. actual success

Team xBA vs Actual Batting Average for 2025 Season

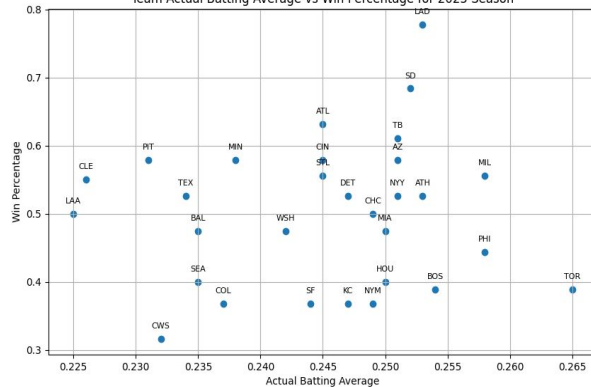


Team xBA vs Win Percentage for 2025 Season



However, these stats do not always serve as comprehensive measures of winning percentage (hence the search for a more predictive metric).

Team Actual Batting Average vs Win Percentage for 2025 Season



Inference/Modeling

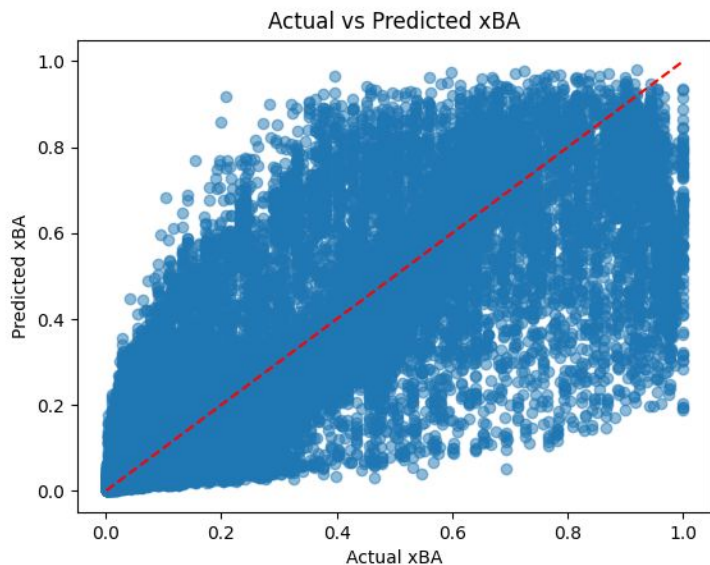
To predict hitting success, we tested 2 different models with slightly different focuses:

1. Linear regression model to predict the probability that a batted ball resulted in a hit (functionally the same as xBA)
2. Logistic regression model to predict a binary outcome of each ball (1=hit, 0=no hit)

Because we only had 7 unique variables that had a significant impact, we used 2nd degree interaction terms in both models to see how the different variables affect one another in predicting the probability of a hit.

Additionally, we tested the logistic regression with a dynamic threshold (default = 0.5) of what should be classified as a predicted hit

Linear Regression Results



MSE = ~ 0.030 , $R^2 = \sim 0.627$.

This is relatively accurate, especially considering the multitude of features our model didn't take into account that can also factor into whether a batted ball results in a hit.

We can also interpret which features were the most heavily weighed in this model by looking at the strongest coefficients:

hit_distance: 3.6980

launch_speed²: 2.9889

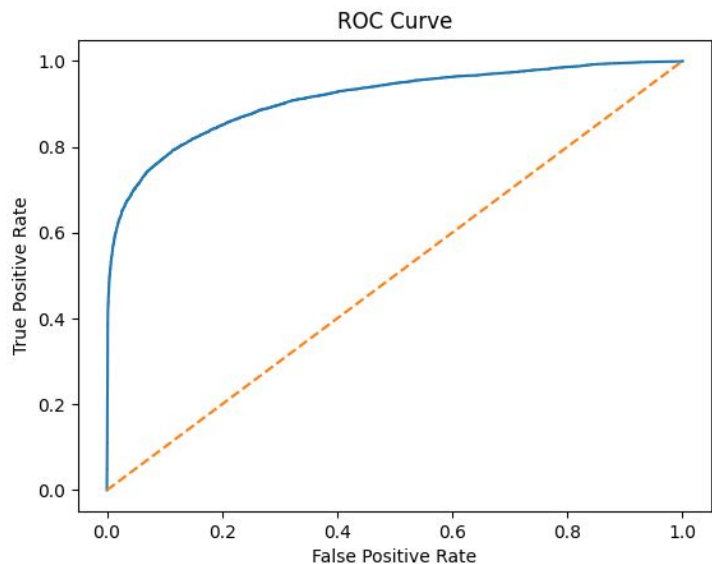
launch_angle * hit_distance: -2.8950

launch_speed: -2.2277

hit_location_x²: 1.5922

From this, we can see that exit velocity heavily impacts predicted xBA, while launch angle does not, despite both being factored into the actual (unknown) formula for xBA.

Logistic Regression Results

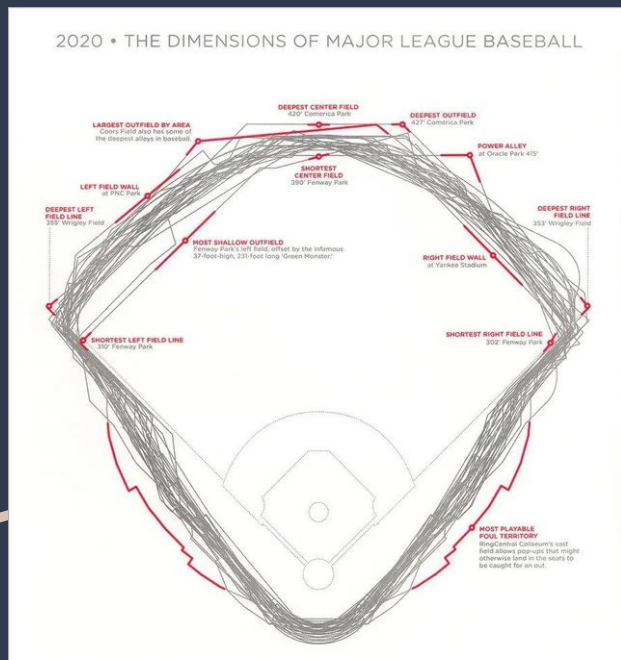


The best logistic regression model used a classification threshold of 0.63, which yielded a confusion matrix of: 15,294 true negatives, 1,190 false positives, 2,023 false negatives, and 5,908 true positives.

Classification Report:				
	precision	recall	f1-score	support
0	0.81	0.99	0.89	16484
1	0.98	0.52	0.68	7931
accuracy			0.84	24415
macro avg	0.89	0.76	0.79	24415
weighted avg	0.86	0.84	0.82	24415
ROC-AUC Score: 0.9117793403803021				

Additionally, an ROC-AUC score of ~0.912 is a measure of significant predictive efficiency - it can correctly rank a randomly chosen hit higher than a randomly chosen non-hit about 91.2% of the time (much more accurate than linear regression model).

Limitations/Next Steps



While our models show that xBA can be consistently predicted and used to confidently quantify hitting success (to a certain extent), there are still limitations with our methods.

Every stadium in baseball has unique dimensions, and while we standardized hit location coordinates, they can be used to create different models for each stadium.

We could also look at defensive stats by team to see who had above or below average success fielding the ball, and how that might change the xBA of opposing teams.

Lastly, our final dataset may have exhibited some survivorship bias since we filtered by players who had at least 50 plate appearances in 2025, decreasing the volatility of season stats at the cost of obscuring our models' ability to predict on less proficient hitters.

